SIEMENS

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What's new in TIA Portal Openness? 2

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TIA Portal Openness: API for automation of engineering workflows

Manual

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Siemens Aktiengesellschaft
Digital Industries
Postfach 48 48
90026 NÜRNBERG
GERMANY

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Cybersecurity information 1

Cybersecurity information

Siemens provides products and solutions with industrial cybersecurity functions that support the secure operation of plants, systems, machines and networks.

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What's new in TIA Portal Openness? 2

Compatibility and long term stability

Siemens.Engineering libraries

The TIA Portal Openness architecture V21 has been extensively redesigned for seamless integration and long-term stability. The result is new modular TIA Portal Openness libraries with simplified assembly management.

Because the Siemens.Engineering libraries of V17, V18, V19, V20 are not included in the scope of delivery, applications and Add-Ins based on V17, V18, V19, V20 must be adapted to run in TIA Portal V21.

The new Siemens.Engineering libraries can be found in the installation directory under "PublicAPI\V21\net48\". For example, the V21 base library can be found in "C:\Program Files\Siemens\Automation\Portal V21\PublicAPI\V21\net48\Siemens.Engineering.Base.dll".

See also: Major changes for long-term stability in TIA Portal Openness V21

Exporting SimaticML files

The Siemens.Engineering libraries in the TIA Portal V21 installation will create SimaticML files of engineering version V21.

Importing SimaticML files

The Siemens.Engineering libraries in the TIA Portal V21 installation can import SimaticML files from engineering versions V18, V19, V20, and V21.

New features

The following new features and innovations are available in TIA Portal Openness V21. For more details on the various topics, refer to the relevant sections of the product documentation.

- General
  - TIA Portal Openness is no longer a setup option but an inherent feature of TIA Portal
  - Improved setup and update performance of products and options
  - Extensive redesign of the TIA Portal Openness architecture with new modular Openness libraries; see also: Major changes for long-term stability in TIA Portal Openness V21
  - Simplified assembly resolving dependent assemblies
  - Version-independent AllowList for the TIA Portal Openness firewall
  - Extended self-description support of APIs
  - Improved formatting of SimaticML
- TIA Portal Add-Ins

  - New modular Openness libraries; see also: Major changes for long-term stability in TIA Portal Openness V21
  - New types of TIA Portal Add-Ins: Corporate Add-Ins, System Add-Ins, and User Add-Ins
  - Create shared .NET class libraries for Openness and Add-Ins with uniformed Openness libraries
  - Possibility to get and attach to other TIA Portal processes
- Projects
  - Device specific export of project texts with category filter
- Libraries
  - Use master copies and library types for PLC documents such as Named value types (NVT)
- Hardware configuration
  - Create and manage communication connections (FDL, HMI, ISO, ISO-on-TCP, PtP, S7, TCP, UDP)
  - Features for hardware configuration in run (CiR)
  - Export PN-GSD file of an I-Device
  - Manage CP / TIM TeleControl data points
  - Configure project integrated shared device
  - Query status of GSDX signature for GSD devices
- Security
  - Support additional confidential configuration data protection settings of a PLC
- Hardware modules
  - Hardware parameter access for S7-1200 G2 (Standard)
  - Bulk operation support for Safety module parameters: ET200 SP Safety, ET200 AL Safety, ET200pro Safety, ET200eco PN Safety, ET200MP Safety
  - Access to further hardware parameters of SCALANCE
  - Resource configuration and export of configuration file for S7-1500 Software Controller (V30.0)
- CAx data exchange
  - Release of CAx Publication Tools V4.0 for hardware data management compatible with TIA Portal V21
- Online scenarios
  - Verify certificate details of a secure connection in online event handlers
  - Check secure communication state for passwords in online event handlers
  - Go online on a protected PLC
  - Go online on a PLC by specifying the IP address

  - Go online on a primary or backup R/H PLC by specifying the IP address
  - Speedup of reading online fingerprints of a PLC for quick station compare
  - Improved error message in case of issues with a secure connection
  - Download additional user files to PLC
- Station Upload
  - Continue upload device as new station for a PLC if a PROFINET IO device cannot be uploaded
- PLC user programs
  - Extended new text-based file exchange format for PLC blocks for programming language LAD, Safety-LAD, FBD, Safety-FBD, SCL and blocks with mixed languages as well as data blocks and PLC data types
  - Create snapshot of actual values for a data block
  - Load start values as actual values for a data block
  - Load snapshot as actual values for a data block
  - Read and write further block attributes such as for information, compilation, and download
  - Rename groups and objects in PLC software
  - Manage write protection of blocks
  - Access multilingual title and comment of objects in PLC software
  - Get PLC tags of a specific device item
- Safety engineering
  - PROFIsafe addressing capability F-BaseID
  - Read F-signatures from Safety Administration Editor
  - Enable and disable F-activation of an F-PLC
- Safety Validation Assistant
  - Create and manage activation tests
- Technology objects
  - Configure technology object SIMATIC Ident
  - Connecting with Superimposing Axes
  - Data block member and interpreter mapping for motion technology objects
- WinCC Unified
  - Create and manage HMI graphic lists
  - Access additional HMI runtime settings
  - New HMI controls: Alarm Line, Process Diagnostic Criteria Analysis, Process Diagnosis Graph Overview, Process Diagnosis Overview, Process Diagnosis PLC Code Viewer

  - New HMI widgets: Alarm Indicator
  - Export and import layout fields of a screen
- Alarming and ProDiag
  - Export and import supervision names and user labels
- SIMATIC Visualization Architect (SiVArc)
  - Configure acquisition mode
  - Configure layout data
  - Upgrade SiVArc definitions
  - Access tag members
- Web server
  - Manage default web pages and web applications
- OPC UA
  - Generate OPC UA nodes during creation of reference namespaces
  - Rename existing OPC UA reference namespace
  - Configure OPCA UA access control for users and roles
  - Dynamic management of OPC UA client certificates
- Version Control Interface (VCI)
  - Support of F-compliant PLC data types, FBD/LAD/SCL and mixed blocks and data blocks in SIMATIC SD format
  - Support of types in the project library: PLC blocks, PLC data types
- SINAMICS Startdrive
  - Access drive item hardware modules and their component names
  - Upload drive parameters

See also

Major changes for long-term stability in TIA Portal Openness V21

See also

Major changes in TIA Portal Openness V16
Major changes for long-term stability in TIA Portal Openness V17
TIA Portal Openness object model
Major changes in TIA Portal Openness V15.1
Major changes in TIA Portal Openness V15
Major changes in V14 SP1
Major changes in V14

Basics

3

3.1 Installation

3.1.1 Requirements for TIA Portal Openness

Requirements for using TIA Openness applications

- A product based on the TIA Portal is installed on the PC, for example, "STEP 7 Professional" or "WinCC Professional".
- The "TIA Portal Openness" is installed on the PC.

  See Installing TIA Portal Openness

Supported Windows operating systems

The following table shows which combinations of Windows operating system, TIA Portal and user application are mutually compatible:

Windows operating system	TIA Portal	User application
64-bit	64-bit	32-bit, 64-bit and "Any CPU"

Requirements for programming TIA Portal Openness applications

- Microsoft Visual Studio 2017 or later with .Net Framework SDK 4.8 and the Windows Classic Desktop package

Necessary user knowledge

- Knowledge as a system engineer
- Advanced knowledge of Microsoft Visual Studio 2017 or later with .Net Framework SDK 4.8
- Advanced knowledge of C# / VB.net and .Net Framework
- User knowledge of the TIA Portal

TIA Portal Openness remoting channels

The TIA Portal Openness remoting channels are registered as type IpcChannel with the “ensureSecurity” parameter set to "false".

Note

You should avoid registering another IpcChannel using a “ensureSecurity” parameter value other than "false" with a priority higher than or equal to “1”.

The IpcChannel is defined with the following attributes:

Attribute	Settings
"name" and "portName"	Set to $”{Process.Name}_{Process.Id}” or $”{Process.Name}_{Process.Id}_{AppDomain.Id}” when registered in an AppDomain other than the application’s default.
“priority”	Set with the default value of “1”.
“typeFilterLevel”	Set to “Full”.
“authorizedGroup”	Set to the NTAccount value string for the built-in user account (i.e. everyone).

See also

Adding users to the "Siemens TIA Openness" user group

3.1.2 Installing TIA Portal Openness

Introduction

TIA Portal Openness is an inherent feature of TIA Portal and is automatically installed.

Result

Depending on the different TIA Portal products and options, additional modular Siemens.Engineering assemblies will be installed.

A local Windows group "Siemens TIA Openness" is being created once during setup.

Note

Administrators must grant individual users the execution of any TIA Portal Openness application in general.

See also: Adding users to the "Siemens TIA Openness" user group

3.1.3 Adding users to the 'Siemens TIA Openness' user group

Introduction

When you install TIA Portal on your computer, the local Windows group "Siemens TIA Openness" is automatically created.

Whenever you try to access a TIA Portal process with your TIA Portal Openness application, TIA Portal verifies that you are a member of the "Siemens TIA Openness" user group, either directly or indirectly by way of another user group. If you are a member of the "Siemens TIA Openness" user group, the TIA Portal Openness application starts and establishes a connection to the TIA Portal instance, else an EngineeringSecurityException will be thrown telling you can’t access this process.

Note

This group membership permits you to execute any TIA Portal Openness application.

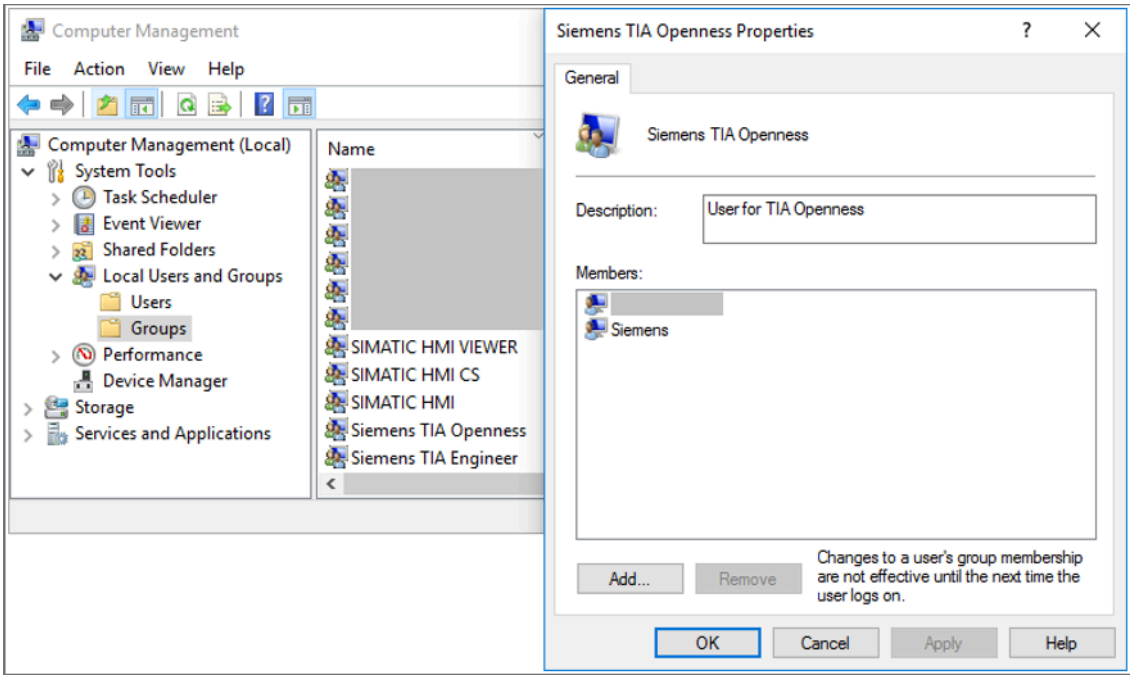
See also: TIA Portal Openness firewall

Procedure

Your administrator must manage the local group membership for "Siemens TIA Openness" in the local Microsoft Windows control panel.

1. Start the local “Computer Management” as administrator from the search in the Windows start menu
2. Go to “Groups” in “Local Users and Groups” to list the local user groups
3. Select “Siemens TIA Openness” and show the properties
4. Click on “Add...” to add a user
5. The affected user must fully sign out and sign in again

Example:



3.1.4 Using Windows Service

Introduction

Windows Services are long-running executables that run without a user interface and are installed using an installation utility. It is possible to develop such a Windows service application that uses TIA Portal Openness.

To do so, it is necessary to create a project of type "Windows Service (.NET Framework)" in Microsoft Visual Studio.

Requirement

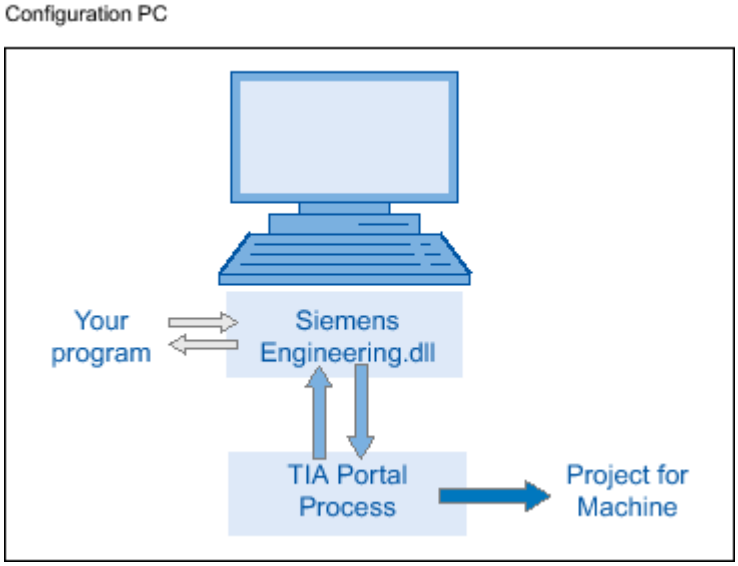
- The process executable of the Windows Service application needs to be added to the AllowList beforehand. This can be done by creating the Windows registry entries on installation of the Windows Service, for example as part of a setup created with a Windows Installer project.
  For more information about the AllowList, see section "Addition of an AllowList entry by connecting to TIA Portal" in chapter TIA Portal Openness firewall
- It is not possible to access running TIA Portal processes or to start TIA Portal with graphical interface. Only a new TIA Portal instance without user interface can be created.
- It is highly recommended to run the Windows Service with credentials of a dedicated Windows user account. It is possible but not recommended to use the local service account or the network service account. The local system account is not supported due to security restrictions and will always throw a EngineeringSecurityException.
- The used Windows account (either service or user) needs to be part of the local Windows user group "Siemens TIA Openness".

See also

TIA Portal Openness firewall

3.1.5 Accessing the TIA Portal

Overview



Procedure

1. Set up the development environment to access and start the TIA Portal.
2. Instantiate the object of the portal application in your program to start the portal.
3. Find the desired project and open it.
4. Access the project data.
5. Close the project and exit the TIA Portal.

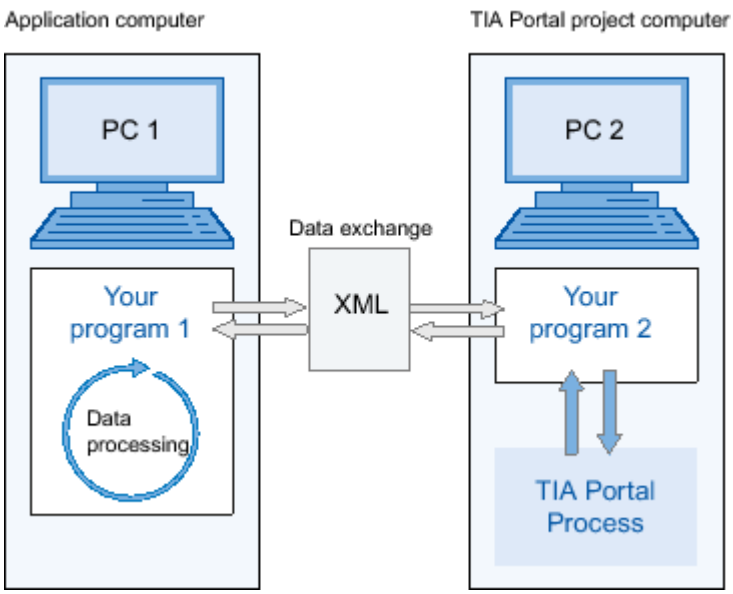
See also

Connecting to the TIA Portal
Terminating the connection to the TIA Portal

3.1.6 Configurations

You can work with two variants of the TIA Portal Openness:

Application and the TIA Portal are on different computers



- Data exchange takes place by XML files. The XML files can be exported or imported by your programs.
- The data exported from the TIA Portal project to PC2 can be modified on PC1 and re-imported.

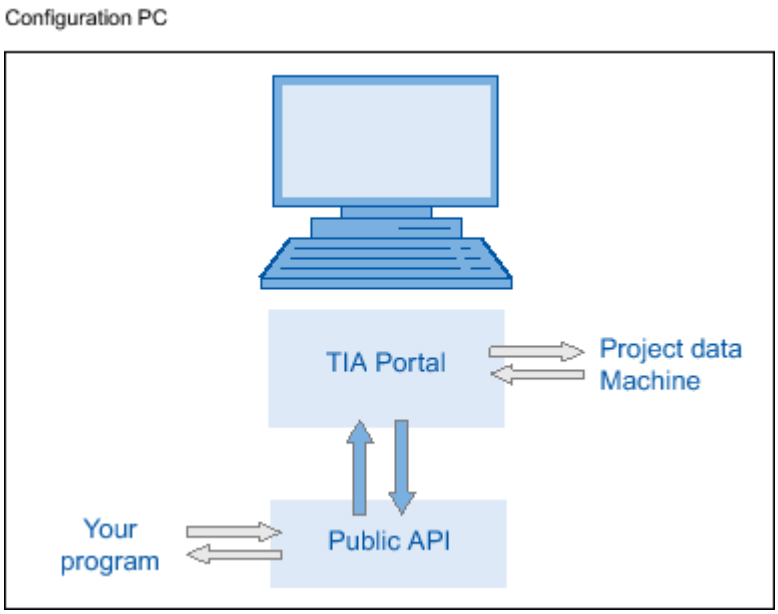
  Note

  You have to develop an executable program "Your Program 2" for PC2, such as "program2.exe". The TIA Portal runs with this program in the background.

  Import and export of XML files takes place exclusively via the TIA Portal Openness API.

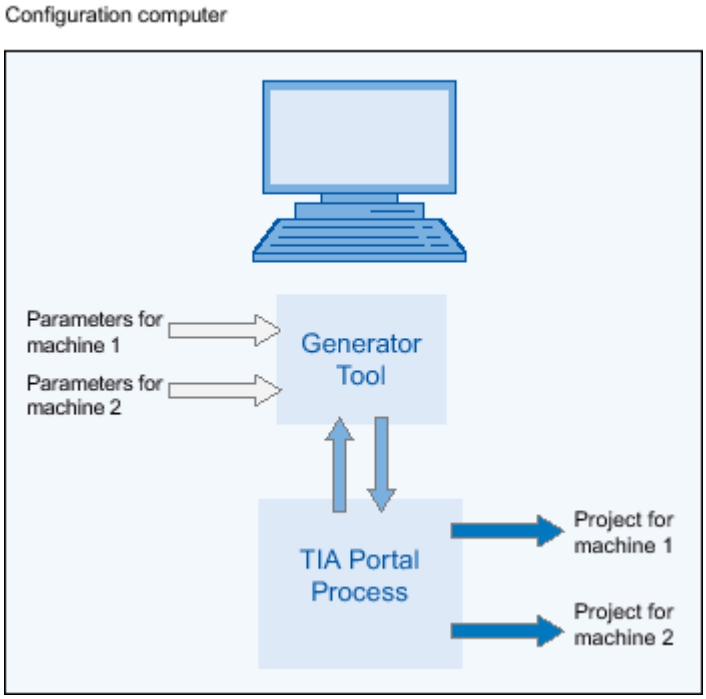
- You can archive exchanged files for verification purposes.
- Exchanged data can be processed at different locations and times.

Application and the TIA Portal are on the same computer



- Your program launches the TIA Portal either with or without the user interface. Your program opens, saves and/or closes a project. The program can also connect to a running TIA Portal.
- You can then use the TIA Portal functionality to request, generate and modify project data or to initiate import or export processes.
- The data is created under control of the TIA Portal processing and stored in the project data.

Typical application in modular mechanical engineering



- An efficient automation system is to be applied to similar machines.
- A project is available in the TIA Portal, which contains the components of all machine variants.

- The Generator tool controls the creation of the project for a specific machine variant.
- The Generator tool obtains the defaults by reading in the parameters for the requested machine variant.
- The Generator tool filters out the relevant elements from the overall TIA Portal project, modifies them if necessary and generates the requested machine project.

3.2 Openness tasks

3.2.1 Introduction

Introduction

TIA Portal Openness describes open interfaces for engineering with the TIA Portal. You will find further information about "TIA Portal Openness - Efficient generation of program code using code generators" in the SIEMENS YouTube channel.

You automate the engineering with TIA Portal Openness by controlling the TIA Portal externally from a program you have created.

You can perform the following actions with TIA Portal Openness:

- Create project data
- Modify projects and project data
- Delete project data
- Read in project data
- Make projects and project data available for other applications.

Note

Siemens is not liable for and does not guarantee the compatibility of the data and information transported via these interfaces with third-party software.

We expressly point out that improper use of the interfaces can result in data loss or production downtimes.

Note

The code snippets contained in this documentation are written in C# syntax.

Due to the shortness of the used code snippets the error handling description has been shortened as well.

Application

The TIA Portal Openness interface is used to do the following:

- Provide project data.

- Access the TIA Portal process.
- Use project data.

Using defaults from the field of automation engineering

- By importing externally generated data
- By remote control of the TIA Portal for generating projects

Providing project data of the TIA Portal for external applications

- By exporting project data

Ensuring competitive advantages through efficient engineering

- You do not have to configure existing engineering data in the TIA Portal.
- Automated engineering processes replace manual engineering.
- Low engineering costs strengthen the bidding position as compared to the competition.

Working together on project data

- Test routines and bulk data processing can take place in parallel to the ongoing configuration.

See also

Configurations

3.2.2 Applications

Introduction

TIA Portal Openness provides you with various ways to access the TIA Portal and offers a selection of functions for defined tasks.

You access the following areas of the TIA Portal by using the TIA Portal Openness API interface :

- Portal data
- Project data
- PLC data
- HMI data
- Drive data

Accessing the TIA Portal

TIA Portal Openness offers various ways to access the TIA Portal. You create an external TIA Portal instance in the process either with or without UI. You can also access ongoing TIA Portal processes at the same time.

Accessing projects and project data

When accessing projects and project data, you mainly use TIA Portal Openness for the following tasks:

- Close, open and save the project
- Enumerate and query objects
- Create objects
- Delete objects

3.2.3 Export/import

Introduction

TIA Portal Openness supports the import and export of project data by means of XML files. The import/export function supports external configuration of existing engineering data. You use this function to make the engineering process effective and free of error.

Application

You use the import/export function for the following purposes:

- Data exchange
- Copying parts of a project
- External processing of configuration data, for example, for bulk data operations using find and replace
- External processing of configuration data for new projects based on existing configurations
- Importing externally-created configuration data, for example, text lists and tags
- Providing project data for external applications